CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 442 تدريب

12/01/2023 · ساعة ونص · 25 درجة

محارب

01 عن الميد

ميد الفصل 442 بتاريخ 12/01/2023 لنفس الدكتور. الأسئلة منقولة من صور ورقة طالب مصححة.

Question	1	2	3	4	5	6	Total
Marks	2	4	6	8	3	2	25

مهم

هذا الميد منهجه مختلف: فيه Heaps و Binary search trees و Graphs، وهذي مو في محاضرات هذا الفصل اللي وصلتني. المفيد منه لك: Q2 (True/False)، و Q3b و Q5 (Hash tables)، و Q6 (عدّ العمليات).

سؤالين ما نقلتهم كاملين لأن شكلهم ما ينقرأ في الصور: شكل الـ graph في Q1، وشكل الشجرة في Q4c. وما عندي نموذج إجابة رسمي، فالحلول من حلي.

الأسئلة 02

BIPARTITE GRAPH · 2 M **Q1**

Check if this graph is bipartite or not, and why?

(The graph figure is not readable in the photo.)

The answer: _____ The reason is: _____

FILL IN + TRUE / FALSE · 2.5 + 1.5 M

Q2

a) Fill in the blanks

• InsertItem to dictionary takes _____ time.

• Bottom-up heap construction runs in _____ time.

• In the binary search tree: Let u, v, and w be three nodes such that u is in the left subtree of v and w is in the right subtree of v. Therefore we have: key(u) ≤ _____ ≤ key(w).

• The space used by a Queue with a Singly Linked list is _____

• FindElement in binary tree takes _____

b) Put T or F for each of the following statements.

( ) In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size.

( ) Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.

( ) The total of primitive operations represents the Worst case of an algorithm.

HEAPS + HASH FUNCTION · 4 + 2 M

Q3

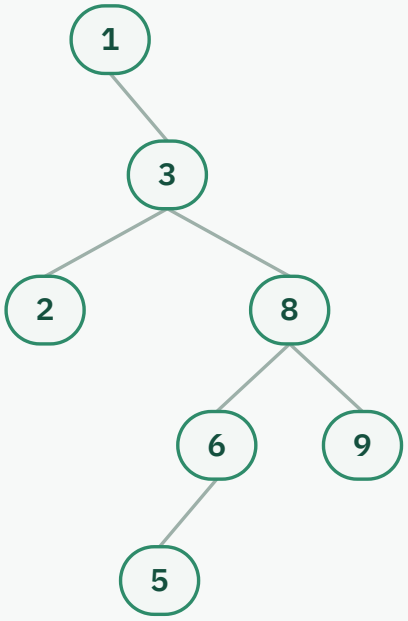
a) what is the objective of Upheap, Downheap algorithms? when to use? then what time required to run UpHeap and DownHeap?

b) When you can say that the hash function is "good"?

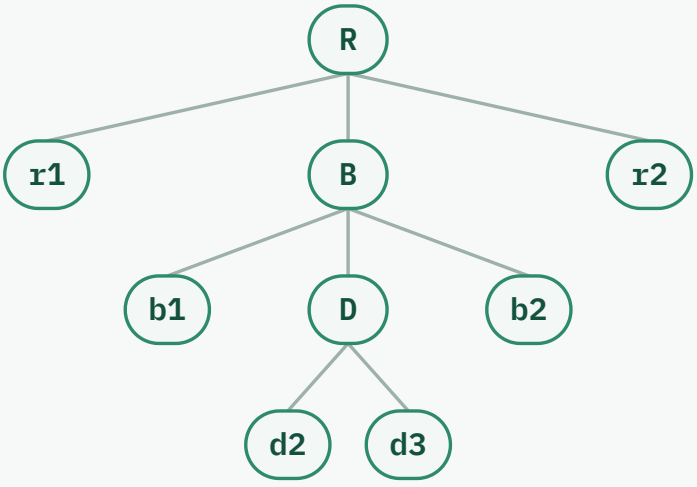
TREES · 2 + 4 + 2 M

Q4

a) Remove the node 3 from the following binary tree, then redraw the tree after deletion.



b) Design the Linked Data Structure that represents the following Tree structure.



c) Write the sequence of tree traversal of the figure. *(The figure is cut off in the photo.)*

COLLISION HANDLING · 3 M

Q5

Discuss all available methods for handling collisions in hash tables.

COUNTING OPERATIONS · 2 M

Q6

Count the total primitive operations of the following algorithm.

```
Algorithm arrayMax(A, n)
    currentMax ← A[0]
    Temp ← A[0]
    for i ← 1 to n − 1 do
        if A[i] > currentMax then
            currentMax ← A[i]
    { increment counter i }
    return currentMax
```